



Giant Cell Tumor of Bone with Atypical Cells in a 6-Year-Old Boy: A Case Report

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INTRODUCTION

- Giant cell tumors (GCTs) of the bone account for **21%** of all benign tumors and **5% of all primary bone tumors**.
- World Health Organization classification → GCT of the bone was redefined from a **benign to intermediate group**.
- GCTs mostly occur at the epi-metaphysis in 11 patients (52.3%)

The vertebral localization is rare (3%)

Kim WJ, Kim S, Choi DW, Lim GH, Jung ST. Characteristics of Giant Cell Tumor of the Bone in Pediatric Patients: Our 18-Year, Single-Center Experience. *Children* (Basel). 2021 Dec 8;8(12):1157

Board W. *Soft Tissue and Bone Tumours: International Agency for Research on Cancer*. IARC; Lyon, France: 2020

Louraoui SM, Fliyou F, Ait Benhamou R, El Azhari A. Destructive Giant Cell Tumor With a Secondary Aneurysmal Bone Cyst of Cervical Spine: A Rare Pediatric Case Report. *Cureus*. 2022 Mar 18;14(3):e23303

INTRODUCTION

Atypical variant is characterized by an abundance of osteoclastic giant cells and a paucity of osteoid tumor, leading it to be confused with giant cell tumors.

Surgical removal is the gold standard

Subsequent covering of the defect by **polymethylmethacrylate** (PMMA) or **resection with joint replacement** are performed.

PRESENTATION OF THE CASE

Anamnesis

- A 6-yo boy
- The patient presented with a lump on the nape of the neck that appeared 6 months ago following a fall while playing soccer. The lump was massaged three times, reducing in size over 2 months. However, it re-enlarged and persisted after another fall from a vehicle 4 months ago, accompanied by night pain and limited range of motion (ROM)

PRESENTATION OF THE CASE

Past Medical History

- Trauma: Positive (6 and 4 months ago).
- Family history: Positive for lumps in the father (hand, back, abdomen), not examined

PRESENTATION OF THE CASE

Physical Examination

Back Region

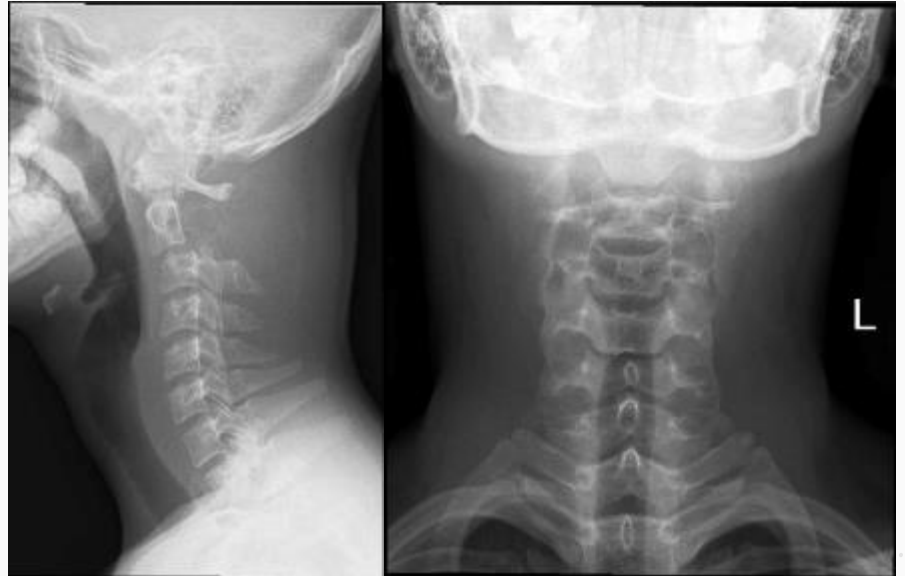
- L : Mass (+), Redness (-)
- F : Tenderness (+), warmth (-), mobile (-), well-defined (-), padat keras, NV distal (+) normal
- M : ROM limited dt mass



PRESENTATION OF THE CASE

Cervical X-ray AP/Lat

- Primary bone tumor suggestive of malignancy in the posterior column of C2 vertebra, suspected osteosarcoma.
- Grade II anterolisthesis of C2 over C3

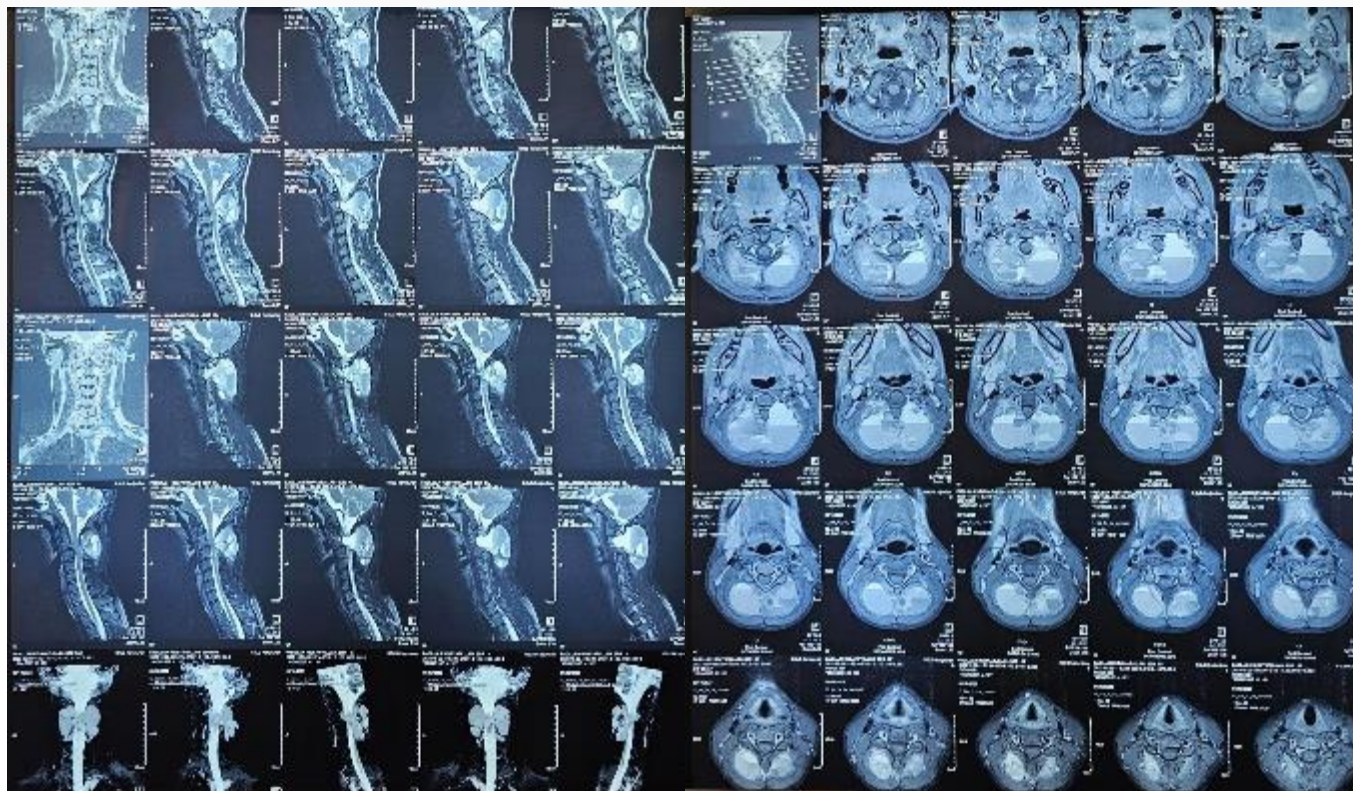


PRESENTATION OF THE CASE

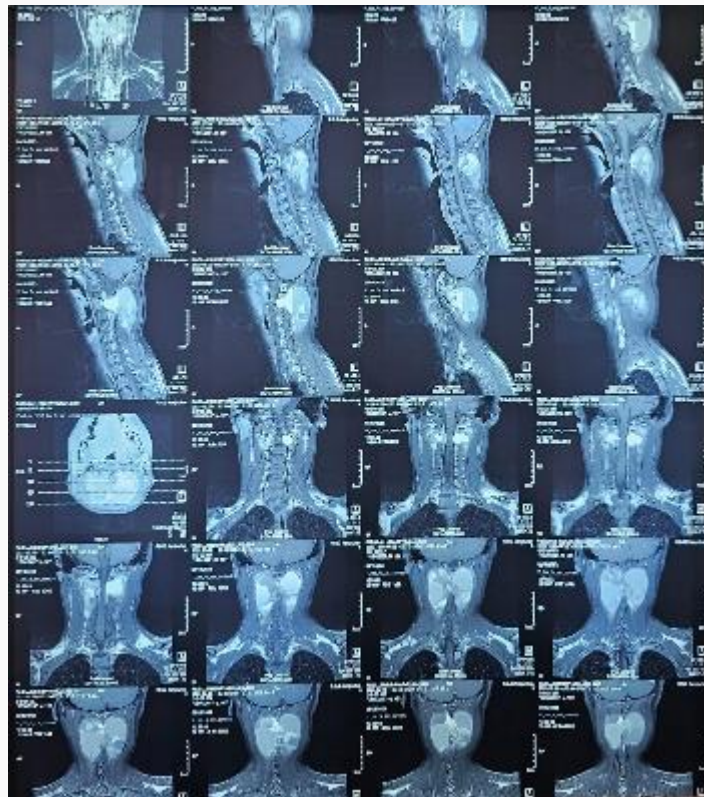
MRI Cervical w/o Contrast

- Primary bone tumor at C2 lamina, bilateral articular and spinous processes, supporting the appearance of an aneurysmal bone cyst (ABC) causing:
 - Grade I anterolisthesis of C2 over C3
 - Moderate spinal canal and foraminal narrowing at C2-C3
- Straight cervical alignment

PRESENTATION OF THE CASE



PRESENTATION OF THE CASE



PRESENTATION OF THE CASE

Fine Needle Aspiration Biopsy

Giant cell-containing tumor suspicious for Giant Cell Tumor of Bone with atypical cells.

PRESENTATION OF THE CASE

Diagnosis

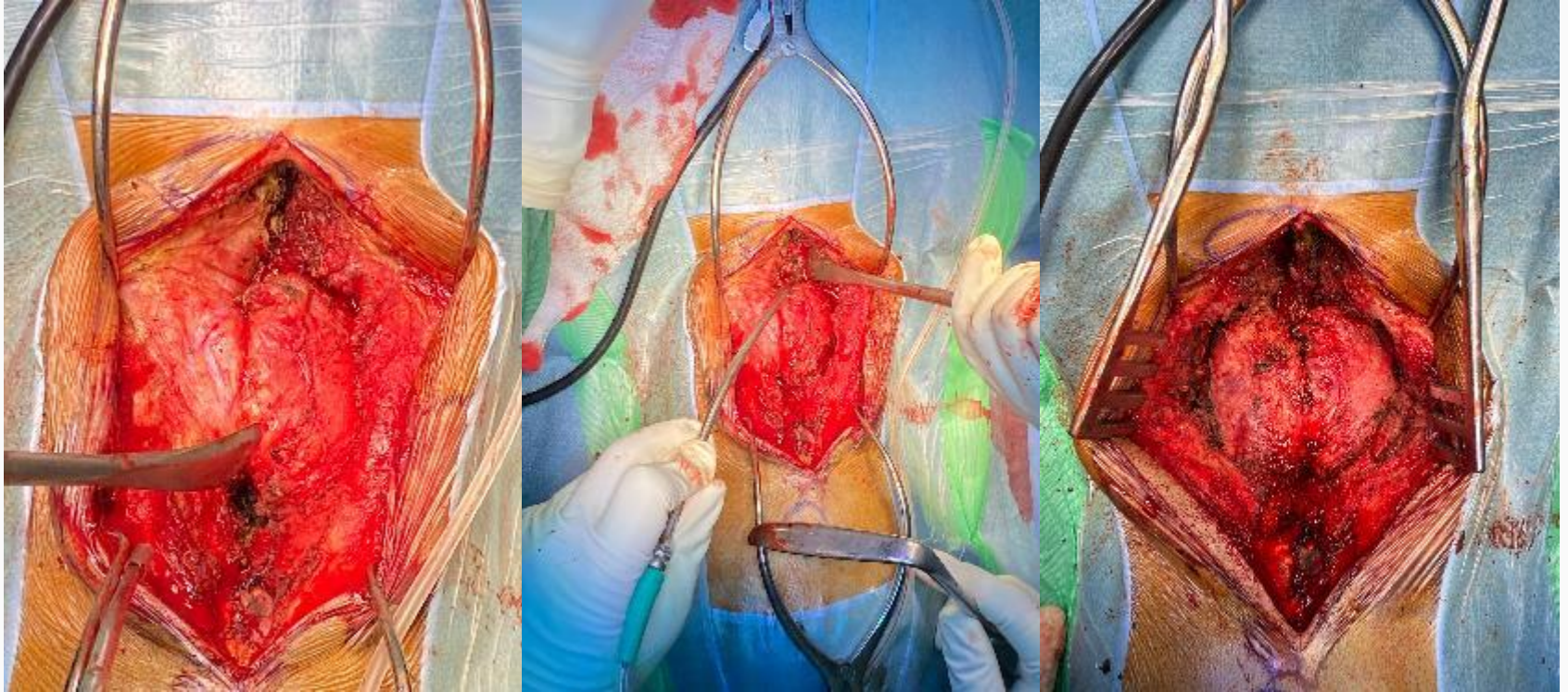
Giant Cell Tumor at Vert C2 Fr.E

PRESENTATION OF THE CASE

Treatment

- Analgetic
- Tumor excision + Occipitocervical Fixation with IOM

Procedure of Surgery



Procedure of Surgery



Procedure of Surgery



PRESENTATION OF THE CASE

Clinical Presentation Post-OP



PRESENTATION OF THE CASE

Cervical X-ray Post-OP

- Absent normal structure of C2 transverse and spinous processes; occipitocervical fixation intact.
- Straight cervical alignment



DISCUSSION

Treatment: Systematic Review

- GCRO is found mostly in children, adolescents, and young adults, with half of the cases analyzed occurring in patients **under 20 years old**
- For effective local tumor management, malignant tumors should be treated with **wide en-bloc resection**, including surrounding intact tissue, without breaching the tumor capsule to minimize recurrence risk.
- **Total en-bloc resections in the cervical spine**, with vertebral artery control or sacrifice, have been previously described.

Clarke MJ, Price DL, Cloft HJ, Segura LG, Hill CA, Browning MB, et al. En bloc resection of a C-1 lateral mass osteosarcoma: technical note. J Neurosurg Pediatr. (2016) 18(1):46–52. 10.3171/2015.12.PEDS15496 [PubMed] [CrossRef] [Google Scholar]

González-Díaz R, Ortiz-Cruz. En-bloc corpectomy for cervical spine tumors by a single anterior approach under vertebral artery control. Technique and two cases report. May 2010. Cervical Spine Research Society.

González-Díaz R, Koller H. Cirugía de la Columna cervical. Madrid: Editorial Panamericana; (2015). ISBN 10: 8498358086

DISCUSSION

Treatment: Systematic Review

- Post-resection, cervical spine stability and alignment were restored using circumferential or **occipitocervical fusion**.
- For the cervicothoracic tumor, a stable reconstruction involved **anterior and posterior spinal fusion**, with tricortical iliac **crest allograft**, titanium Moss mesh, anterior plate, and rods with varying screw diameters (**4.5 mm cervical, 5.5 mm thoracic**).

DISCUSSION

Treatment

- Delangel (2023) states that the best treatment for spinal neoplasms is to completely remove the tumor to prevent recurrence and fully decompress the spinal canal, which alleviates neurological symptoms. Surgical treatment can be supplemented with radiotherapy to lower the risk of recurrence

CONCLUSION

- For a 6-year-old boy with a Giant Cell Tumor of Bone with atypical cells, effective treatment involves **complete tumor removal with wide en-bloc resection**, including surrounding tissue, to reduce recurrence risk.
- Post-surgery, **cervical spine stability** is restored through **fusion techniques**, and radiotherapy can be used to further decrease recurrence risk.